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Class :- SYIT	Batch :- B1
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Practical 30

Code -

```
1  import breeze.linalg._
2  object prac30 extends App {
3    // Dataset
4    val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)
5
6    // Initial cluster centers
7    var center1 = 2.0
8    var center2 = 10.0
9
10   // Repeat clustering
11   for (i <- 1 to 5) {
12     val cluster1 = data.toArray.filter(x =>
13       | math.abs(x - center1) < math.abs(x - center2)
14     )
15
16     val cluster2 = data.toArray.filter(x =>
17       | math.abs(x - center2) <= math.abs(x - center1)
18     )
19     // Calculate new centers
20     center1 = cluster1.sum / cluster1.length
21     center2 = cluster2.sum / cluster2.length
22   }
```

```

23 // Display clusters
24 println("Cluster 1:")
25 println(data.toArray.filter(x =>
26 |   math.abs(x - center1) < math.abs(x - center2)
27 | ).mkString(", "))
28
29 println("Cluster 2:")
30 println(data.toArray.filter(x =>
31 |   math.abs(x - center2) <= math.abs(x - center1)
32 | ).mkString(", "))
33
34 println("Center 1 = " + center1)
35 println("Center 2 = " + center2)
36
37 println("Jasad S024")
38 }
39

```

Output :-

```

↑ "C:\Program Files\Java\jdk1.8.0_181\
↓
Cluster 1:
1.0, 2.0, 3.0
Cluster 2:
10.0, 11.0, 12.0
Center 1 = 2.0
Center 2 = 11.0
Jasad S024

Process finished with exit code 0
|

```